



Smart Health

DISTRICT OPS

AI-driven health centre and supply monitoring

TRACK 3 — SMART HEALTH

Real-time stock monitoring, multilingual reporting, and AI-driven redistribution
for PHCs and CHCs district-wide

THE PROBLEM

Shortages are discovered after they've already become a crisis

PHCs and CHCs across a district currently track medicine stock, patient footfall, bed availability, and doctor attendance manually. There is no real-time visibility into any of it — which means a shortage, an absent doctor, or a full ward is only discovered once a patient is already standing in front of it.

No real-time visibility

Stock, beds, and attendance tracked on paper or spreadsheets updated late

Reactive, not proactive

Shortages surface only once a patient is turned away

Language barriers

Manual reporting assumes English or a single regional language

THE SOLUTION

Report in any language. Let AI do the analysis.

Smart Health District Ops lets field staff report issues in their own language — Hindi, Hinglish, Kannada, or English — with zero technical effort. Two AI agents then do the work a district administrator would otherwise do by hand: turning messy text into clean data, and turning raw stock numbers into a ranked, actionable recommendation.

Built to be maintained

Runs entirely on Google's ecosystem, so a real health department can operate it without a dedicated engineering team.

System architecture



HOW IT WORKS

From a voice note to a redistribution plan

01

Field staff report

A message is sent in the worker's own language, or entered via a simple mobile form.

02

AI parses the input

Centre, medicine, quantity, and urgency are extracted automatically — no manual translation.

03

AI analyzes district-wide stock

Every centre's stock is scanned to flag shortages and check for surplus elsewhere.

04

Admins get a ranked action

Not just a warning — a specific recommendation, like which centre to redistribute from.

TECHNOLOGIES USED

Built entirely on tools a district health department can maintain

- **Google Sheets**
Core database — Centres, StockLogs, Alerts, RawInput
- **Google Apps Script**
Backend automation and AI orchestration
- **AI language model (Llama 3.3 70B)**
Multilingual parsing and stock-risk analysis
- **Google AppSheet**
Mobile field data-entry application
- **Looker Studio**
Live, color-coded admin dashboard
- **Google Sites**
Public-facing project landing page
- **Gemini (Nano Banana)**
Logo generation
- **GitHub**
Source code hosting and documentation

SCOPE

What's fully built vs. what's next

Fully built and working

- Multilingual stock reporting (AI parsed)
- Stock-out risk detection
- AI redistribution recommendations
- Live admin dashboard

Natural next extensions

- Bed availability wired into AI alerts
- Doctor attendance wired into AI alerts
- Patient footfall tracking
- Test-availability audits

Centre-level bed and doctor-attendance data already exist in the database — the same two-agent architecture already proven for stock extends directly to them.

SEE IT LIVE

Links

- **Project site**
sites.google.com/view/smart-health-districtops/home
- **Live dashboard (Looker Studio)**
datastudio.google.com/reporting/882a62e6-1432-4e93-ad99-1bca37a37299
- **Mobile data-entry app (AppSheet)**
appsheet.com/newshortcut/33e691a7-3bb0-4698-b9ca-8099f127b94c
- **Demo video (YouTube)**
youtu.be/53WsbikIbPI
- **GitHub repository**
github.com/adiithi0510-alt/Smart-health-district-ops